

some noise on T_e and the input currents, and that the disturbance on i_1 caused a steady-state error in x_3 .

This result can be modified by changing the estimator. Let us suppose that we wish to reduce the steady-state error on x_3 and we can tolerate increased noise effects.¹⁹ So let us increase the assumption for the process noise Eq. (9.129) to

$$\mathbf{R}_w = \begin{bmatrix} 0.001 & 0 \\ 0 & 0.001 \end{bmatrix}.$$

By increasing the process noise, we are essentially saying that the knowledge of the plant model is less precise; therefore, the optimal filter will pay more attention to the measurements by increasing the estimator gains and speeding up the estimator. Higher estimator gains should reduce the steady-state estimation errors and their effect on x_3 , but at the cost of increased sensitivity to measurement noise. Carrying out the calculations yields

$$\mathbf{L} = \begin{bmatrix} 0.497 & -0.143 \\ 0.497 & 0.143 \\ -0.338 & 0.329 \\ 0.338 & 0.329 \end{bmatrix}, \quad (9.132)$$

which is larger in every term than the previous gain in Eq. (9.130). Using the gain to form the estimator error equation and transforming its roots to the s -plane results in

$$s = -4.0 \pm j8.2, -2.1 \pm j20.6 \text{ sec}^{-1},$$

which are faster than the previous roots in Eq. (9.131). Evaluation of the same case as in Fig. 9.19 with everything identical except for the estimator gains results in Fig. 9.20. Note that the actual magnitude of the measurement noise and disturbance was *not* changed in generating the figure. The value of $\mathbf{R}_w$ was changed in order to produce higher values in the gain matrix. The figure shows the expected result: larger sensitivity to the measurement noise but reduced sensitivity to the disturbance.

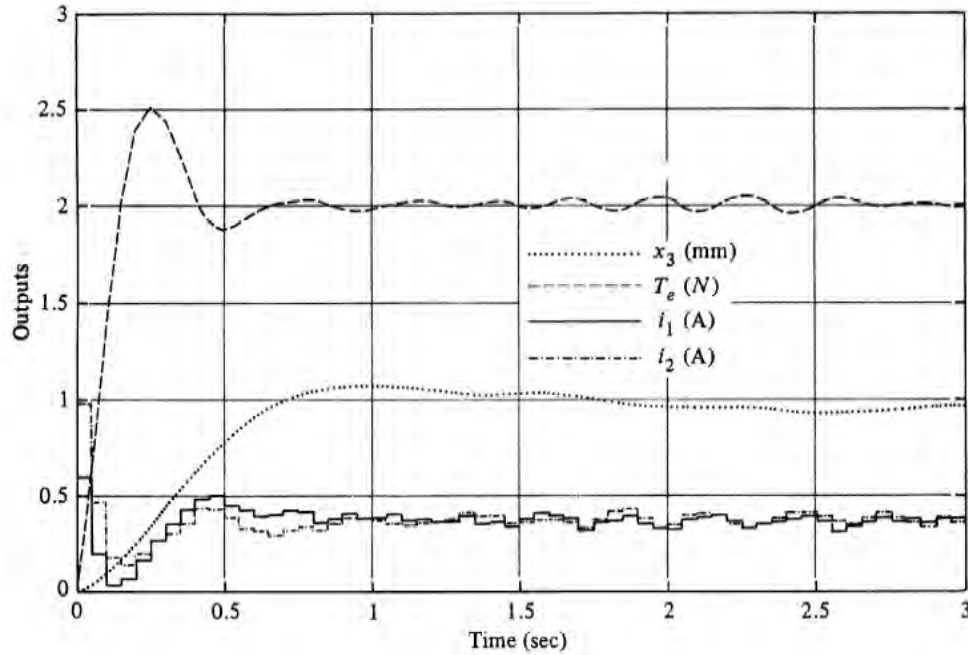
Decoupled Design

As a final comment, we note that the plant can be uncoupled and designed as two independent second-order systems, one for x_3 and one for T_e . The key idea to accomplish the uncoupling is to note that equal commands on the two current inputs will cause an increase in tension with no change in x_3 , whereas equal and

¹⁹ We could also add a second integral control loop on x_3 to kill the error, but it is preferable to reduce errors with high control and estimator gains in order to have a robust system for all kinds of errors, if the gains can be tolerated.

Figure 9.20

Time response with integral control and an estimator with L from Eq. (9.132) including measurement noise and a disturbance at 1.5 sec



opposite current commands will cause a change in x_3 with no effect on tension. We therefore transform the control accordingly

$$\mathbf{u}_d = \begin{bmatrix} u_3 \\ u_t \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} \Rightarrow \mathbf{T}_u = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}. \quad (9.133)$$

We also define a new state

$$\mathbf{x}_d = \begin{bmatrix} x_3 \\ \dot{x}_3 \\ T^e \\ \dot{T}^e \end{bmatrix} = \begin{bmatrix} \mathbf{H}_3 \\ \mathbf{H}_3 \mathbf{F} \\ \mathbf{H}_t \\ \mathbf{H}_t \mathbf{F} \end{bmatrix} \mathbf{x} = \mathbf{T}_d \mathbf{x}, \quad (9.134)$$

where $\mathbf{H}_3$ and $\mathbf{H}_t$ are the partitions of $\mathbf{H}_e$ in Eq. (9.127) and $\mathbf{F}$ is the system matrix from Eq. (9.117). Following the state transformation ideas in Section 4.3.3, we can write that

$$\dot{\mathbf{x}}_d = \mathbf{F}_d \mathbf{x}_d + \mathbf{G}_d \mathbf{u}_d, \quad (9.135)$$

where

$$\mathbf{F}_d = \mathbf{T}_d \mathbf{F} \mathbf{T}_d^{-1} \quad \text{and} \quad \mathbf{G}_d = \mathbf{T}_d \mathbf{G} \mathbf{T}_u^{-1}. \quad (9.136)$$

Performing these calculations yields

$$\mathbf{F}_d = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -66.3 & -1.18 \end{bmatrix} \quad \text{and} \quad \mathbf{G}_d = \begin{bmatrix} 0 & 0 \\ 42.7 & 0 \\ 0 & 3.2 \\ 0 & 176.5 \end{bmatrix}. \quad (9.137)$$

Therefore, we see that there is a decoupling that results in two separate systems,

$$\begin{bmatrix} \dot{x}_3 \\ \ddot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ \dot{x}_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 42.7 \end{bmatrix} u_3, \quad (9.138)$$

and

$$\begin{bmatrix} \dot{T}_e \\ \ddot{T}_e \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -66.3 & -1.18 \end{bmatrix} \begin{bmatrix} T_e \\ \dot{T}_e \end{bmatrix} + \begin{bmatrix} 3.2 \\ 176.5 \end{bmatrix} u_t. \quad (9.139)$$

The two system equations, (9.138) and (9.139), can then be used to design two control systems using SISO methods. The resulting controls, u_3 and u_t , can then be “unscrambled” via $\mathbf{T}_u$ to arrive at the desired i_1 and i_2 , that is,

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \mathbf{T}_u^{-1} \begin{bmatrix} u_3 \\ u_t \end{bmatrix}.$$

The perfect uncoupling was due to the perfect symmetry of the problem: motors and capstan inertias identical, motor constants identical, and tape read head exactly in the middle of the capstans. In an actual system, there would be differences and some coupling would be present. Due to the simplicity of the uncoupled design and its implementation, however, it can be worthwhile to evaluate the performance of a decoupled design to see if its errors warranted the full MIMO design.

9.6 Summary

- An optimal controller is one that minimizes a quadratic cost function where weighting matrices are selected by the designer to achieve a “good” design. Criteria for evaluation of the design are based on all the same factors that have been discussed throughout the book.
- Optimal control methods are useful in designing SISO systems because they help the designer place the pole locations.
- Optimal control methods are essential in designing MIMO systems as they reduce the degrees of freedom in the design iteration down to a manageable few.

- Generally, for linear, stationary systems, the steady-state optimal control solution is satisfactory and far more practical than the true optimal time-varying gain solution.
- The most accurate discrete emulation method is based on finding the discrete equivalent of a continuous cost function and solving for the discrete optimal control system with that cost.
- An optimal estimator is based on achieving the minimum mean square estimation error for a given level of process and measurement noise acting on the system.
- Generally, for linear, stationary systems, the steady-state optimal estimator is satisfactory and far more practical than the true optimal time-varying gain solution (Kalman filter).
- Optimal estimators are useful for SISO system design and essential for MIMO system design.
- The optimization procedures provide a technique for which many good design candidates can be generated which can then be evaluated to ascertain whether they meet all the design goals. The procedures do not eliminate trial-and-error methods; rather, they transfer iteration on classical compensation parameters to iteration on optimal cost-function parameters or assumed system noise characteristics.

9.7 Problems

9.1 Optimal control derivation:

- Derive Eq. (9.33) and hence verify the control Hamiltonian as given by Eq. (9.34).
- Demonstrate that if $\mathbf{W}$ is a transformation that brings $\mathcal{H}_c$ to diagonal form, then the first n columns of $\mathbf{W}$ are eigenvectors of $\mathcal{H}_c$ associated with stable eigenvalues.
- Demonstrate that if the optimal steady-state control, $\mathbf{K}_\infty$, given by Eq. (9.47) is used in the plant equation Eq. (9.10), then the matrix in the upper left corner of $\mathbf{W}$, which is $\mathbf{X}_l$, has columns that are eigenvectors of the closed-loop system.

9.2 Symmetric root locus:

- Compute the closed-loop pole locations of the satellite design problem, using the steady-state gains given by Fig. 9.3.
- Take the discrete transfer function from u to θ for the satellite design problem (Appendix A) and form the symmetric root locus. Plot the locus and locate the values of ρ corresponding to the selections of Q_2 given in Eq. (9.30).
- Show that for a system with one control it is always possible to construct a symmetric root locus corresponding to the optimal steady-state control. [Hint: Show that if $\mathbf{Q}_1 = \mathbf{H}\mathbf{H}^T$ and $\mathbf{G}(z)$ is a column-matrix transfer function given by $\mathbf{H}(z\mathbf{I} - \Phi)^{-1}\Gamma$, then the roots of the optimal control are given by $1 + \rho\mathbf{G}^T(z^{-1})\mathbf{G}(z) = 0$, which is a scalar symmetric root-locus problem and which can be put in the form $1 + \rho G_1(z^{-1})G_1(z) = 0$. Use Eq. (9.38).]

- (d) Give the equivalent scalar plant transfer function $G_1(z)$, if we have the satellite control problem and use $\mathbf{Q}_1 = \mathbf{I}$, the 2×2 identity. Draw the symmetric root locus for this case.
- 9.3 Compute the location of the closed-loop poles of the optimal filter for the satellite problem for each of the values of R_w given by the steady final values of the curves of Fig. 9.8.
- 9.4 For the antenna example in Appendix A.2,
- Design an optimal controller corresponding to the cost function $\mathcal{J}(y, u) = u^2 + \rho y^2$ for $\rho = 0.01, 0.1$, and 1 . Use $T = 0.2$ sec. Plot the step response of the closed loop system for initial errors in y and $\dot{y}$ in each case. Which value of ρ most nearly meets the step-response specification given in Chapter 7?
 - Draw the symmetric root locus corresponding to the design of part (a).
 - Design an optimal steady-state filter for the antenna. Assume that the receiver noise has a variance $R_v = 10^{-6} \text{ rad}^2$ and that the wind gust noise, w_d , is white, and we want three cases corresponding to $R_{wpsd} = 10^{-2}, 10^{-4}$, and 10^{-6} .
 - Plot the step response of the complete system with control law corresponding to $\rho = 0.1$ and filter corresponding to $R_{wpsd} = 10^{-4}$.
- 9.5 The lateral equations of motion for the B-767 in cruise are approximately

$$\begin{bmatrix} \dot{v} \\ \dot{r} \\ \dot{\phi}_p \\ \dot{\rho} \end{bmatrix} = \begin{bmatrix} -0.0868 & -1 & -0.0391 & 0 \\ 2.14 & -0.228 & 0 & -0.0204 \\ 0 & 0 & 0 & 1 \\ -4.41 & 0.334 & 0 & -1.181 \end{bmatrix} \begin{bmatrix} v \\ r \\ \phi_p \\ \rho \end{bmatrix} + \begin{bmatrix} 0.0222 & 0 \\ -1.165 & -0.065 \\ 0 & 0 \\ 0.549 & -2.11 \end{bmatrix} \begin{bmatrix} \delta_r \\ \delta_a \end{bmatrix}$$

- Design two second-order controllers by ignoring the cross-coupling terms. Pick roots at $s = -1 \pm j 1.5$ rad/sec for the yaw mode (v and r) and $s = -2, -0.1$ for the roll mode (ϕ_p and ρ). Use a sample period of $T = 0.05$ sec.
 - Determine the root locations that result when the cross-coupling terms are included. (Use `eig.m` in MATLAB.)
 - Assuming one can measure r and ϕ_p , design two second-order estimators by ignoring the cross-coupling terms. Place all poles at $s = -2$ rad/sec.
 - Determine the root locations that result from (c) when the cross-coupling terms are included.
- 9.6 The equations of motion for a stick balancer (Fig. 9.2) are given by

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \\ \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 21 & 0 & 0 & 0.8 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -0.4 \end{bmatrix} \begin{bmatrix} \theta \\ \omega \\ x \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \end{bmatrix} u,$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \omega \\ x \\ v \end{bmatrix}.$$

Design two second-order estimators, one for the stick (θ, ω) and one for the cart (x, v). Verify that the one-way coupling (cart to stick only) causes the full fourth-order estimator to have the same roots as the separate second-order designs.

- 9.7 Starting with the paper-machine design in Section 9.5.3 whose output is shown in Fig. 9.11, perform design iterations on Q_1 , Q_2 , and t_s to find the lowest possible value of u_1 that still meets the state specifications on maximum value and settling time. [Hint: The routines FIG99.M, FIG910.M and FIG911.M from the Digital Control Toolbox will be useful to you.]
- 9.8 For the double mass-spring system in Example 9.5, investigate the effect of different Q 's on the closed-loop poles of the system. In particular, determine a Q that will yield an equivalent damping of the oscillatory poles that is greater than $\zeta = 0.5$. [Hint: FIG95.M from the Digital Control Toolbox will be useful to you.]
- 9.9 For the double mass-spring system in Example 9.5, design a controller and estimator that uses a measurement of d . It should respond to a command input for d with a rise time of 2 sec with an overshoot less than 20%.
- 9.10 For the double mass-spring system in Example 9.5, design a controller and estimator that uses a measurement of d . It should respond to a command input for d with a rise time of 3 sec with an overshoot less than 15%. Plot the frequency response of your compensation (control plus estimator) and qualitatively describe the features of the compensation.
- 9.11 A simplified model of a disk drive is

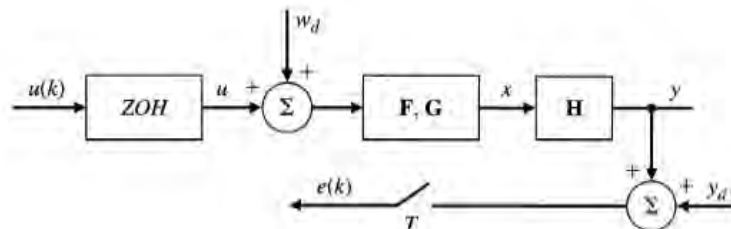
$$G(s) = \frac{10^6 \omega_r^2}{s^2(s^2 + 2\zeta \omega_r s + \omega_r^2)},$$

where $\omega_r = 19,000$ rad/sec ($\cong 3$ KHz) and $\zeta = 0.05$. This is the transfer function between the torque applied to the disk head arm and the motion of the read head on the end of the arm. Note this is also the same generic model as the double mass-spring system in Eq. (A.20). The system has the capability to measure where the heads are with respect to the data tracks on the disk. The two biggest disturbances are due to an actuator bias (w_d) and track wobble (y_d) from an off center disk mounting. A schematic diagram of the system to be controlled is shown in Fig. 9.21.

Since the disk is rotating at 3822 rpm (400 rad/sec), an offset causes a sinusoidal y_d to be added to the output of the transfer function above. The purpose of the servo is to follow the data tracks as closely as possible, whether the disk is slightly off center or not because that allows higher data densities on the disk. It is also desirable for the servo to settle on a new track as quickly as possible because the disk access time is directly related to this settling time and is a major selling point for disk drives.

Design a discrete compensator with a sample rate no faster than 8 kHz that has an overshoot to a step command no greater than 20%, does not allow steady state tracking

Figure 9.21
Disk drive closed-loop
system for Problems
9.11 and 9.12



errors due to the actuator bias, and has a settling time (to within 5% of the final value) to a step input less than 5 msec. The system should be robust with respect to variations in the resonance frequency as this can change by $\pm 2\%$ due to temperature changes. Plot the closed-loop frequency response of the *error*, e , in order to evaluate how well your system attenuates the disturbance response in the vicinity of 400 rad/sec (3822 rpm).

- 9.12** Design a compensator for the disk drive servo specified in Problem 9.11 so that it provides the minimum possible error due to the disturbance at 3822 rpm. Since the spin rate drifts some, define the error to be the average of that from: the nominal spin rate of 3822 rpm, $+1\%$ of the nominal, and -1% of the nominal. (*Hint*: Review Section 8.5.)

- 9.13** For

$$G(s) = \frac{1}{s^2},$$

use optimal control and estimation to design compensation that provides a 10 msec rise time with less than a 10% overshoot to a command. Determine the closed-loop system bandwidth and the system phase margin.

- 9.14** The MATLAB code below will generate some noisy data from an oscillator driven by a square wave

```
F=[0 1;-1 0];G=[0;1];H=[1 0];J=0;T=0.5;
sysC = ss(F,G,H,J)
sysD=c2d(sysC,T)
Un=[ones(1,10) zeros(1,10) ones(1,10) zeros(1,11)];
U=Un+0.1*randn(size(Un));
Yn=lsim(sysD,U);
Ym=Yn+0.2*randn(size(Yn)).
```

The measurement noise is zero mean and normally distributed with $R_v = 0.2^2$ while the process noise is zero mean and normally distributed with $R_w = 0.1^2$. Assume the initial value of the state has a covariance matrix

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and use a time-varying Kalman filter to reconstruct the state. Compare the resulting time history of the optimal state estimate with that from a constant gain Kalman filter. [*Hint*: FIG98.M from the Digital Control Toolbox will be useful to you.]

- 9.15** Repeat Example 9.7, but fit the best straight line to the data. [*Hint*: FIG97.M from the Digital Control Toolbox will be useful to you.]

♦ 10 ♦

Quantization Effects

A Perspective on Quantization

In Chapter 13 we will consider the analysis of systems having general nonlinearities; but first, in this chapter, we will consider the special nonlinearity of digital control: Numbers in the computer must be forced to fit in digital words that are defined by a finite number of bits, usually 8, 16, 32, or 64 bits. Thus far we have considered the use of the digital computer as a linear, discrete-time device for implementing control designs. Now we turn to consideration of the fact that numbers in the computer are taken in, stored, calculated, and put out with finite accuracy. The technique of representing a real number by a digital value of finite accuracy affects digital control in two principal ways. First, the variable values such as e , u , and the internal state variables, $\mathbf{x}$, used in the difference equations are not exact, and thus errors are introduced into the output of the computer. As we shall see, these errors can often be analyzed as if there were noise sources in the computer. Second, the coefficients such as a_i and b_i of the difference equations, which must be stored in the computer, cannot be arbitrary real numbers but must be realizable by a finite number of bits. This can cause the machine to solve a slightly different equation than it was designed to do and has the potential to result in instability. Our purpose in this chapter is to explore methods that can be used for the analysis of these two effects: quantization of variables and quantization of coefficient parameters. The effect of quantization when the microprocessor has 32 bits is typically not noticeable. However, some products are very cost sensitive and it is desirable to use an 8 bit computer, if possible. The material in this chapter is primarily directed toward the design of such systems.

Chapter Overview

The random error model for the round-off process is first developed in Section 10.1. This section also develops several analysis methods for such random processes that are useful for quantization and then applies the random process

analysis methods from Chapter 9 to the quantization case. Section 10.2 examines the deterministic problem of parameter storage quantization errors and demonstrates the sensitivity of the errors to the structure of the difference equation mechanization. The last section, 10.3, describes some of the consequences of round-off errors and what steps can be taken to alleviate them.

10.1 Analysis of Round-Off Error

fixed point

floating point

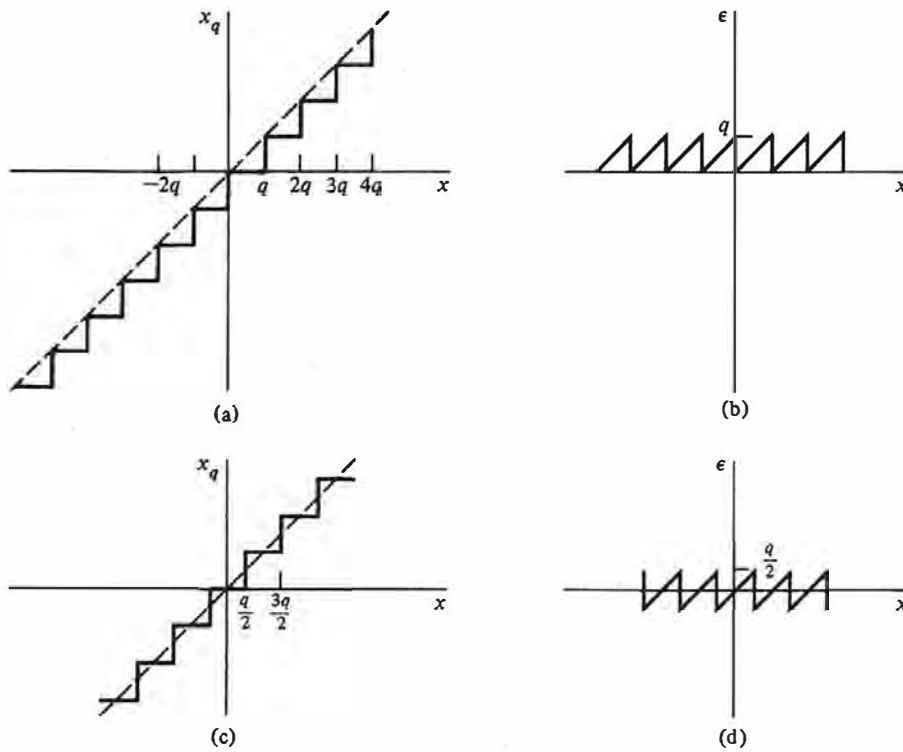
quantization

round-off

In our first analysis of finite accuracy we will assume that the computer represents each number with a fixed location for the equivalent of the decimal point, or **fixed point** representation. Although for any given number the point is fixed, considerations of amplitude scaling and dynamic range often require that different numbers in the same program have their decimal points in different locations. The consequence is that different magnitudes of errors are introduced at different locations as the results of calculations being fitted into the proper computer number format. Fixed point representation is typical in real-time control computers; however, in scientific calculations done in higher-level languages such as BASIC, MATLAB, or C, **floating point** representations are mainly used, wherein the number representation has both a mantissa (magnitude and sign information) and an exponent that causes the location of the decimal point to float. With fixed-point arithmetic, addition and subtraction are done without error except that the sum can overflow the limits of the representation. Overflow must be avoided by proper amplitude scaling or analyzed as a major nonlinearity. For control- or signal-processing applications, the logic is arranged so that the result of overflow is a saturation effect, a nonlinearity that will be treated in Chapter 13. In the case of multiplication, however, a double-length product is produced and the machine must reduce the number of bits in the product to fit it into the standard word size, a process we generally call **quantization**. One way to perform quantization is by ignoring the least significant half of the product, a process called **truncation**. If we assume that the numbers are represented with base 2 and that ℓ binary digits (bits) are to the right of the point (for the fractional part), the least significant bit kept represents the magnitude $2^{-\ell}$. The result of truncation will be accurate to this value (the part thrown away could be almost as large as $2^{-\ell}$). Thus a plot of the "true" value of a variable x versus the quantized value, x_q , would look like Fig. 10.1(a), where the error (shown in (b)) is decided by the quantum size, q (which is $2^{-\ell}$ under the conditions mentioned above). It is common practice in control computers to use **round-off** rather than truncation. With round-off, the result is the same as truncation if the first bit lost is a 0, but the result is increased by $2^{-\ell}$ if the first bit lost is a 1. The process is the same as is common with ordinary base 10 numbers where, for example, 5.125 is rounded to 5.13, but 5.124 becomes 5.12 to two (decimal) places of accuracy. An input-output plot of rounding is shown in Fig. 10.1(c), and the corresponding error is shown in

Figure 10.1

Plot of effects of number truncation. (a) Plot of variable versus truncated values. (b) Plot of error due to truncation. (c) Plot of variable versus rounded values. (d) Round-off error



10.1(d).¹ Notice that the maximum error with roundoff is half that resulting from truncation. The value of ℓ in a particular case depends on the word size of the particular computer in use. For control implementations the choice is based on the required accuracy and dynamic range and is limited by the expense. At this time (1997), microprocessors are readily available with word sizes of 8, 16, and 32 bits, with a cost premium on the larger sizes. One of the goals of this chapter is to analyze the effect of ℓ and thus of word size on the stability and on the errors due to quantization's effects so the designer can select the smallest word size consistent with the required performance.

We can represent either truncation or round-off by the equation

$$x = x_q + \epsilon, \quad (10.1)$$

¹ The plots in Fig. 10.1 assume that the numbers are represented in two's complement code.

where ϵ is the error caused by the truncation or round-off of x into the digital representation x_q . Because the error due to truncation equals a constant plus round-off error, we will assume round-off in our analysis unless otherwise stated and assume that analysis of the effects of the additional constant bias when truncation is used (which is rare) can be treated separately. Clearly the process of analog-to-digital conversion also introduces a similar effect, although often with still another value for q than those resulting from round-off during arithmetic in the microprocessor.

quantization models

The analysis of the effects of round-off depends on the model we take for ϵ . We will analyze three such models, which we can classify as (a) worst case, for which we will bound the error due to round-off; (b) steady-state worst case, for which we will compute the largest output possible if the system reaches a constant steady state; and (c) stochastic, for which we will present a model of the round-off error as a random process and compute the root-mean-square of the output error due to round-off.

The Worst-Case Error Bound

Bertram bound

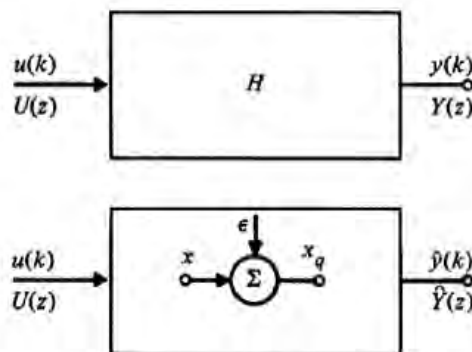
The worst-case analysis is due to Bertram (1958). His analysis takes the pessimistic view that the round-off occurs in such a way as to cause the maximum harm, and his analysis bounds the maximum error that can possibly result from round-off. First we consider the case shown in Fig. 10.2, in which a single case of quantization is assumed to occur somewhere in an otherwise linear constant system.

There is a transfer function from the point of round-off to the output, which we will call $H_1(z)$, and we can describe the situation by the equations

$$\begin{aligned} Y(z) &= H(z)U(z), \\ \hat{Y}(z) &= H(z)U(z) - H_1(z)E_1(z; x), \\ Y - \hat{Y} &= \tilde{Y} = H_1(z)E_1(z; x). \end{aligned} \quad (10.2)$$

Figure 10.2

A linear system and the introduction of one source of round-off errors



We write E_1 as a function of the state variable x to emphasize that Eq. (10.2) is *not* linear because we do not know how to compute E_1 until we have the values of x . However, we are not looking for the exact value of $\tilde{Y}$ but an upper bound on the time error, $\tilde{y}(k)$.

Because we wish to find a bound on the time signal $\tilde{y}(k)$, we need the time domain equivalent of Eq. (10.2), which is the convolution sum given by

$$\tilde{y}(n) = \sum_{k=0}^n h_1(k) \epsilon_1(n-k; x). \quad (10.3)$$

If we examine Fig. 10.1(d), we can see that whatever the exact values of ϵ_1 may be, its magnitude is bounded by $q_1/2$, where q_1 is the quantum value for the operation which introduces the quantization being analyzed. We can use this information to determine in short steps the bound on $\tilde{y}$ as we did in Chapter 4 in the discussion of BIBO (bounded input, bounded output) stability. Taking the magnitudes of both sides on Eq. (10.3), we get

$$|\tilde{y}| = \left| \sum_0^n h_1 \epsilon_1 \right|.$$

Because the sum is bounded by the sum of the magnitudes of each term, we have the inequality

$$\leq \sum_0^n |h_1 \epsilon_1|,$$

which is the same as

$$\leq \sum_0^n |h_1| |\epsilon_1|;$$

but by Fig. 10.1(d), the error magnitude is always less than $q_1/2$, so the output error is bounded by

$$\leq \sum_0^n |h_1| \frac{q_1}{2}.$$

Finally, the sum can only get larger if we increase the number of terms

$$|\tilde{y}(n)| \leq \sum_0^\infty |h_1| \frac{q_1}{2}. \quad (10.4)$$

Equation (10.4) is Bertram's worst-case bound. The function `qwc.m` computes it. By comparison with the condition for BIBO stability, we can conclude that if the linear system is BIBO stable in response to inputs applied at the point of the quantization, then introduction of the quantization will not cause the system to be

unstable in the BIBO sense.³ As we will see later, the system with quantization can have an output error that is nonzero either as a constant or as an oscillation; so the system may not be asymptotically stable but the output error will not grow beyond the bound given by Eq. (10.4).

◆ Example 10.1 *Bertram Bound Application*

For the first-order system

$$y(k+1) = \alpha y(k) + u(k), \quad (10.5)$$

compute Bertram's bound, assuming that the round-off occurs in the computation of the product αy .

Solution. Using Eq. (10.4), we have

$$\hat{y}(k+1) = \alpha \hat{y}(k) + \epsilon(k) + u(k), \quad (10.6)$$

and thus

$$\tilde{y}(k+1) = \alpha \tilde{y}(k) - \epsilon(k). \quad (10.7)$$

For the error system, the unit pulse response is $h_1(k) = \alpha^k$, and Eq. (10.4) is easily computed as follows for $|\alpha| < 1$

$$|\tilde{y}| \leq \frac{q}{2} \sum_{k=0}^{\infty} |\alpha|^k \leq \frac{q}{2} \frac{1}{1-|\alpha|}. \quad (10.8)$$

For this example, if $\alpha > 0$, the error due to quantization at any particular time is bounded by the *DC* gain from the quantizer to the output times $q/2$. This bound is only likely to be approached when a system has a constant input and has settled to its steady-state value.

The Steady-State Worst Case

The steady-state worst case was analyzed by Slaughter (1964) in the context of digital control, and by Blackman (1965) in the context of digital filters. For this analysis, we view the round-off to cause some stable, transient errors of no special concern, and we assume that all variables eventually, in the steady state, become constants. We wish to know how large this steady-state error can be as a result of round-off. We consider again the situation shown in Fig. 10.2 and thus Eq. (10.3). In this case, however, we assume that Eq. (10.3) reaches a steady state,

³ We assume that there is no saturation of the quantizer during the transient.

at which time ϵ is constant and in the range $-q/2 \leq \epsilon_{ss} \leq q/2$. Then Eq. (10.3) reduces to

$$\tilde{y}_{ss}(\infty) = \sum_0^{\infty} h_1(n) \epsilon_{ss}.$$

The worst steady-state error is the magnitude of this signal with $\epsilon_{ss} = q/2$, which is

$$|\tilde{y}_{ss}(\infty)| \leq \left| \sum_0^{\infty} h_1(n) \right| \frac{q}{2}. \quad (10.9)$$

There is one nice thing about Eq. (10.9): The sum is the value of the transfer function $H_1(z)$ at $z = 1$, and we can write⁴

$$|\tilde{y}_{ss}(\infty)| \leq |H_1(1)| \frac{q}{2}. \quad (10.10)$$

◆ Example 10.2 Worst Case Steady-State Bound

Use the worst case analysis to compute the error bound for the first-order system given by Eq. (10.5).

Solution. In this case, the transfer function is

$$H_1(z) = 1/(z - \alpha).$$

From Eq. (10.10), we find that the bound is

$$|\tilde{y}_{ss}| \leq \frac{q}{2} \frac{1}{1 - \alpha}. \quad (10.11)$$

This is the same as the Bertram worst-case bound if $\alpha > 0$.

So we see that this steady-state worst case yields the general result that the error is bounded by $q/2 \times$ dc gain from the quantization source to the output.

Equations (10.4) and (10.9) or (10.10) express the consequences of round-off from one source. For multiple sources, these equations can be extended in an

⁴ A function to compute the quantization steady-state error is given by `qss.m`.

obvious way. For example, if we have K sources of round-off, each with possibly different quantization levels q_j , then Eq. (10.4) becomes

$$\begin{aligned} |\tilde{y}| &\leq \left\{ \sum_{n=0}^{\infty} |h_1(n)| \frac{q_1}{2} + \sum_{n=0}^{\infty} |h_2(n)| \frac{q_2}{2} + \cdots \right\} \\ &\leq \sum_{j=1}^K \sum_{n=0}^{\infty} |h_j(n)| \frac{q_j}{2}. \end{aligned} \quad (10.12)$$

If all the quanta should be equal, the error bound is

$$\begin{aligned} |\tilde{y}| &\leq \left\{ \sum_{n=0}^{\infty} |h_1(n)| + \sum_{n=0}^{\infty} |h_2(n)| + \cdots \right\} \frac{q}{2} \\ &\leq \sum_{j=1}^K \sum_{n=0}^{\infty} |h_j(n)| \frac{q}{2}. \end{aligned}$$

And likewise for multiple sources, Eq. (10.10) is extended to

$$|y_{ss}(\infty)| \leq [|H_1(1)| \frac{q_1}{2} + |H_2(1)| \frac{q_2}{2} + \cdots + |H_K(1)| \frac{q_K}{2}]. \quad (10.13)$$

It is easily possible to express Eq. (10.10) in terms of a state-variable formulation of the equations of motion. Suppose the equations from the point of the quantization to the output are given by

$$\begin{aligned} \mathbf{x}(k+1) &= \Phi \mathbf{x}(k) + \Gamma_1 \epsilon_1(k), \\ \tilde{y}(k) &= \mathbf{H} \mathbf{x}(k) + J \epsilon_1(k). \end{aligned} \quad (10.14)$$

The assumptions of the steady-state worst-case error are that $\mathbf{x}(k+1) = \mathbf{x}(k) = \mathbf{x}_{ss}$ and $\epsilon_1(k) = \epsilon_{1,ss}$. Then

$$\mathbf{x}_{ss} = \Phi \mathbf{x}_{ss} + \Gamma_1 \epsilon_{1,ss}, \quad \tilde{y} = \mathbf{H} \mathbf{x}_{ss} + J \epsilon_{1,ss}. \quad (10.15)$$

Solving for $\tilde{y}$, we find

$$\tilde{y} = [\mathbf{H}[\mathbf{I} - \Phi]^{-1} \Gamma_1 + J] \epsilon_{1,ss},$$

which is bounded by

$$|\tilde{y}| \leq \|[\mathbf{H}[\mathbf{I} - \Phi]^{-1} \Gamma_1 + J]\| \frac{q_1}{2}. \quad (10.16)$$

The major advantage of the steady-state result is the vastly simpler forms of Eqs. (10.13) and (10.16) as compared to (10.12). Unfortunately, Eq. (10.13) does not always hold because of the assumption of a constant quantization error of $q/2$ in the steady state,⁵ and yet the worst-case upper bound given by Eq. (10.12) is often excessively pessimistic. However, in some cases there is great expense

⁵ Note that Eqs. (10.13) and (10.16) are *not* bounds on the error. They do, however, give a valuable estimate on error size, which is easy to compute for complex systems.

due to distortion of the data should a signal overflow, and the absolute bound can be used to select q so that no overflow is possible. An example of the use of Bertram's bound in the design of a spectrum analyzer is given in Schmidt (1978).

Stochastic Analysis of Round-Off Error

The third model for round-off error is that of a stochastic variable. The analysis will follow Widrow (1956) and has two parts: development of a stochastic model for $\epsilon(k)$ and analysis of the response of a linear system to a stochastic process that has the given characteristics. Because the development of the model requires use of somewhat sophisticated concepts of stochastic processes, and because the model can be given a very reasonable heuristic justification without this mathematical apparatus, we develop the model heuristically and proceed with an analysis of the response. A review of the necessary facts from the theory of probability and stochastic processes is to be found in Appendix D.

First, then, we give a heuristic argument for a stochastic model of round-off error. We begin with examination of Fig. 10.1, where we have a plot of the output versus the input of the round-off operation and a sketch of error versus amplitude of the input signal x . If we imagine that $x(n)$ is a random variable that takes values at successive sampling instants in a scattered way across the scale of values, then it seems reasonable to suppose that the sequence of errors $\epsilon(n)$ would be scattered over the entire range of possible values, which is to say, over the range from $-q/2$ to $q/2$. Furthermore, because the "teeth" in the saw like plot of ϵ versus x are linear and contain no flat places that would signal a preference for one value of ϵ over another, it seems reasonable that the values of $\epsilon(n)$ are equally likely to be anywhere in the range $q/2 \leq \epsilon \leq q/2$. Furthermore, if the signal into the quantizer typically moves several quanta during one sample period, it seems reasonable to assume that the values of error at one sample time will not be correlated with errors at other times; that is, its spectrum would be expected to be flat, which we characterize as "white."⁶ The reflection of this argument in terms of stochastic processes is to assert that we can model $\epsilon(k)$ as a white random process having a uniform probability density from $-q/2$ to $q/2$. A plot of the uniform density is shown in Fig. 10.3.

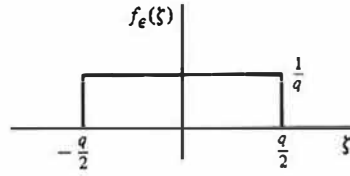
From this density we can immediately compute the mean and variance of ϵ as follows

quantization variance

$$\mu_{\epsilon} = \mathcal{E}\{\epsilon\} = \int_{-\infty}^{\infty} \xi f_{\epsilon}(\xi) d\xi = \int_{-q/2}^{q/2} \xi \frac{1}{q} d\xi = \frac{1}{q} \left. \frac{\xi^2}{2} \right|_{-q/2}^{q/2} = 0, \quad (10.17)$$

⁶ If the input to the quantizer is a deterministic signal such as a square wave, a constant, or a sine wave, this argument is clearly wrong. It is also approximate if the input is a random signal with strong correlation over a time equal to a sampling period. A careful analysis is given in Clavier, et al. (1947). If the input is a sine wave, the error spectrum typically has a number of large spikes, far from flat or white.

Figure 10.3
Plot of the uniform
density function



and

$$\begin{aligned}
 \sigma_\epsilon^2 &= \mathcal{E}\{(\epsilon - \mu_\epsilon)^2\} = \int_{-\infty}^{\infty} (\xi - 0)^2 f_\epsilon(\xi) d\xi = \int_{-q/2}^{q/2} (\xi)^2 \frac{1}{q} d\xi \\
 &= \frac{1}{q} \left. \frac{(\xi)^3}{3} \right|_{-q/2}^{q/2} = \frac{1}{q} \left[\frac{1}{3} (q/2)^3 - \frac{1}{3} (-q/2)^3 \right] \\
 &= \frac{1}{3q} \left[\frac{q^3}{8} + \frac{q^3}{8} \right] = \frac{q^2}{12}.
 \end{aligned} \tag{10.18}$$

Thus we assume the following white-noise model for $\epsilon(n)$ for the case of round-off error as sketched in Fig. 10.1(d)

$$\epsilon(n) = w(n), \tag{10.19}$$

where $w(n)$ has the autocorrelation function (defined in Appendix D)

$$\begin{aligned}
 R_w(n) &= \mathcal{E}\{w(k)w(k+n)\} = q^2/12 \quad (n = 0) \\
 &= 0 \quad (n \neq 0),
 \end{aligned} \tag{10.20}$$

and the mean value μ_ϵ is zero.

With the model given by Eqs. (10.19) and (10.20) we can compute the mean and variance of the system error due to round-off using state-space methods. Again, we observe that analysis of the response to any constant nonzero mean μ_ϵ can be computed separately from the response to the white-noise component $w(n)$.

In order to analyze the effect of the random part of the quantization noise, let's review (see also Section 9.5) the general problem of finding the effect of zero-mean noise, w , on a linear system

$$\begin{aligned}
 \mathbf{x}(k+1) &= \Phi_1 \mathbf{x}(k) + \Gamma_1 w(k) \\
 \tilde{y}(k) &= \mathbf{H}_1 \mathbf{x}(k) + J_1 w(k).
 \end{aligned} \tag{10.21}$$

We define the covariance of the state as

$$\mathbf{R}_x(k) = \mathcal{E}\{\mathbf{x}(k)\mathbf{x}^T(k)\}, \tag{10.22}$$

and, at $k+1$,

$$\begin{aligned}
 \mathbf{R}_x(k+1) &= \mathcal{E}\{\mathbf{x}(k+1)\mathbf{x}^T(k+1)\} \\
 &= \mathcal{E}\{(\Phi_1 \mathbf{x}(k) + \Gamma_1 w(k))(\Phi_1 \mathbf{x}(k) + \Gamma_1 w(k))^T\} \\
 &= \Phi_1 \mathbf{R}_x(k) \Phi_1^T + \Gamma_1 R_w(k) \Gamma_1^T.
 \end{aligned} \tag{10.23}$$

covariance propagation

Equation (10.23) can be used with an initial value (typically zero) of $\mathbf{R}_x$ to compute the transient development of state covariance toward the steady state. Typically, we are only interested in the steady state, which is obtained by letting $\mathbf{R}_x(k+1) = \mathbf{R}_x(k) = \mathbf{R}_x(\infty)$. In this case, Eq. (10.23) reduces to the equation (called the discrete Lyapunov equation for its occurrence in Lyapunov's stability studies)

Lyapunov equation

$$\mathbf{R}_x(\infty) = \Phi_1 \mathbf{R}_x(\infty) \Phi_1^T + \Gamma_1 R_w \Gamma_1^T. \quad (10.24)$$

Several numerical methods for the solution of Eq. (10.24) have been developed, some based on solving Eq. (10.23) until $\mathbf{R}_x$ no longer changes, and others based on converting Eq. (10.24) into a set of linear equations in the coefficients of $\mathbf{R}_x$ and solving these equations by numerical linear algebra as done by `dlyap.m` in MATLAB.

◆ Example 10.3 *Random Quantization Error for a First-Order System*

Determine the output of the first-order system of Eq. (10.5) for the case where the multiplication, $\alpha x(k)$, is rounded with a quantum level q .

Solution. The first-order system of Eq. (10.5), with the rounding error represented by ϵ is

$$x(k+1) = \alpha x(k) + \epsilon(k),$$

and thus

$$\Phi = \alpha \quad \text{and} \quad \Gamma_1 = 1$$

and from Eq. (10.18), we see that

$$R_w = \frac{q^2}{12},$$

so that Eq. (10.24) is

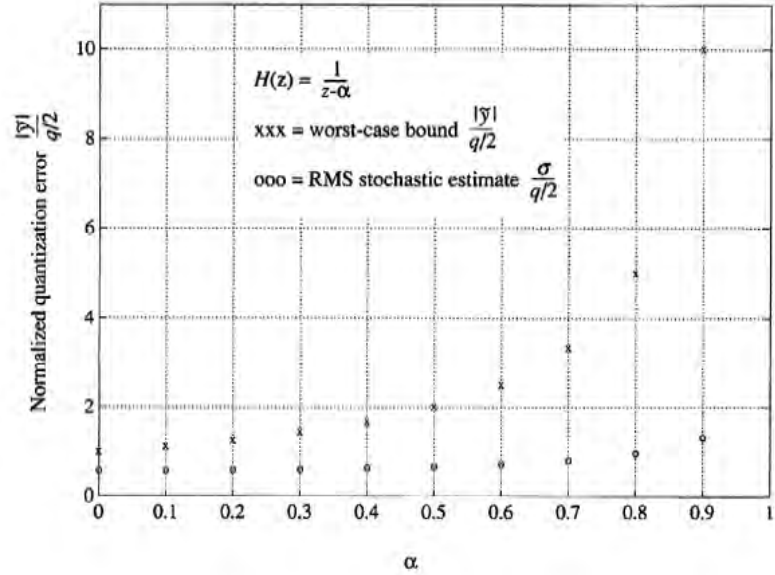
$$\begin{aligned} R_x &= \alpha R_x \alpha + (1) \frac{q^2}{12} (1), \\ R_x &= \frac{q^2}{12} \frac{1}{1 - \alpha^2} \end{aligned} \quad (10.25)$$

Taking the square root, we find that

$$\sigma_x = \sqrt{R_x} = \frac{q}{2} \sqrt{\frac{1}{3(1 - \alpha^2)}}. \quad (10.26)$$

A plot of Eq. (10.26) along with the value of the worst-case bound from Eqs. (10.8) and (10.11) is shown in Fig. 10.4.

Figure 10.4
Error estimates and error
bounds



In some cases, the output of a system is not one of the state elements. In other words,

$$\tilde{y} = \mathbf{H}_1 \mathbf{x} + J_1 w,$$

and, therefore,

$$\mathcal{E}(\tilde{y}\tilde{y}^T) = \mathcal{E}\{(\mathbf{H}_1 \mathbf{x} + J_1 w)(\mathbf{H}_1 \mathbf{x} + J_1 w)^T\},$$

which is to say

$$\mathbf{R}_y = \mathbf{H}_1 \mathbf{R}_x \mathbf{H}_1^T + J_1 \mathbf{R}_w J_1^T. \quad (10.27)$$

Note that we can use Eq. (10.23) and (10.24) to compute the covariance of the state due to round-off error at several locations simply by taking $\mathbf{w}$ to be a column matrix and $\mathbf{R}_w$ to be a square diagonal matrix of covariances of the components of $\mathbf{w}$. In the multiple-source case, $\mathbf{\Gamma}_1$ is a matrix of dimension $n \times p$, where there are n states and p sources.

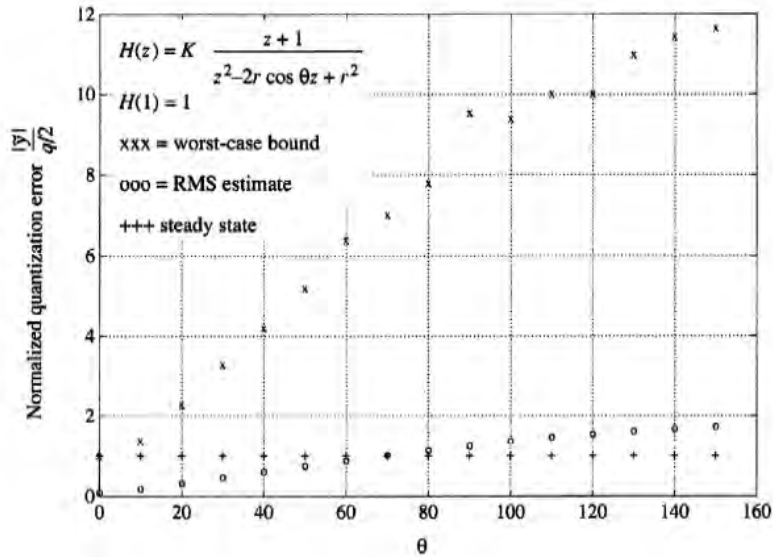
◆ Example 10.4 Random Quantization Error for a Second-Order System

Determine the quantization error of the second-order system

$$y(k+2) + a_1 y(k+1) + a_2 y(k) = \frac{1+a_1+a_2}{2}(u(k) + u(k-1)),$$

in terms of the random root-mean-square response and compare it with the worst-case bounds.

Figure 10.5
Error estimates and error bound with quantization input to a second-order system, where $r = 0.9$



Solution. The gain of the system has been set to be unity at frequency 0, and the parameters are related to the pole locations according to $a_1 = -2r \cos(\theta)$ and $a_2 = r^2$. In order to study the role of the frequency of the pole, given in normalized terms by the angle θ , we have computed the estimated errors according to Eqs. (10.4), (10.16), and (10.27) for several values of θ and plotted them in Fig 10.5. For the random-noise case, the square root of the result of Eq. (10.27) is plotted. From the plot, it is clear that the worst-case bound is quite large compared to the other two computations for most of the range of angles. Experiments in the laboratory with random inputs to the system give results that compare very favorably with the numbers given by the random model.

10.2 Effects of Parameter Round-Off

We have thus far analyzed the effects of round-off on the variables such as y and u in Eq. (10.5). However, to do the calculations for a controller, the computer must also store the equation coefficients, and if the machine uses fixed point arithmetic, the parameter values must also be truncated or rounded off to the accuracy of the machine. This means that although we might design the program to solve

$$y(k+1) = \alpha y(k) + u(k),$$

it will actually solve

$$\hat{y}(k+1) = (\alpha + \delta\alpha)\hat{y}(k) + u(k) + \epsilon(k). \quad (10.28)$$

In this section, we give methods for the analysis of the effects of the parameter error $\delta\alpha$.

The principal concern with parameter variations is that the dynamic response and especially the stability of the system will be altered when the parameters are altered. One way to study this problem is to look at the characteristic equation and ask what effect a parameter change has on the characteristic roots. For example, for the first-order system described by Eq. (10.28), the perturbed characteristic equation is

$$z - (\alpha + \delta\alpha) = 0;$$

and it is immediately obvious that if we want a pole at $z = 0.995$, it will be necessary to store α to three decimal places and that the limit on $\delta\alpha$ for stability is 0.005 because a variation of this magnitude results in a pole on the unit circle. Note, however, that if we structure the α term as $\alpha = 1 - \beta$ and add the 1-term separately then $\beta = 1 - \alpha = 0.005$, and the relative accuracy requirements on β are much less than those on α . Thus we see that the details of the architecture of a realization can have a major impact on the robustness of the design to parameter value quantization. It is also the case that different canonical forms realized with a given parameter accuracy (word size) will be able to realize a different set of pole locations. It is often possible to select the realization structure in such a way that the desired dynamics are almost exactly realized with the available word length.

To study these matters, we consider the characteristic equation and ask how a particular root changes when a particular parameter changes. The general study of root variation with parameter change is the root locus; however, we can obtain results of some value by a linearized sensitivity analysis. We can compare the direct and the cascade realizations of Chapter 4, for example, to see which is to be preferred from a sensitivity point of view. In the direct realization, the characteristic equation is

$$z^n + \alpha_1 z^{n-1} + \cdots + \alpha_n = 0. \quad (10.29)$$

This equation has roots at $\lambda_1, \lambda_2, \dots, \lambda_n$, where $\lambda_i = r_i e^{j\theta_i}$. We assume that one of the α 's, say α_k , is subject to error $\delta\alpha_k$, and we wish to compute the effect this has on λ_j and especially the effect on r_j so that stability can be checked. For this purpose, we can conveniently write Eq. (10.29) as a polynomial $P(z, \alpha)$ that depends on z and α_k . At $z = \lambda_j$, the polynomial is zero, so we have

$$P(\lambda_j, \alpha_k) = 0. \quad (10.30)$$

If α_k is changed to $\alpha_k + \delta\alpha_k$, then λ_j also changes, and the new polynomial is

$$\begin{aligned} P(\lambda_j + \delta\lambda_j, \alpha_k + \delta\alpha_k) &= P(\lambda_j, \alpha_k) + \left. \frac{\partial P}{\partial z} \right|_{z=\lambda_j} \delta\lambda_j + \frac{\partial P}{\partial \alpha_k} \delta\alpha_k + \cdots \\ &= 0, \end{aligned} \quad (10.31)$$

root sensitivity

where the dots represent terms of higher order in $\delta\lambda$ and $\delta\alpha$. By Eq. (10.30) we see that the first term on the right-hand side of Eq. (10.31) is zero. If $\delta\lambda_j$ and $\delta\alpha_k$ are both small, then the higher-order terms are also negligible. Thus the change in λ_j is given to first order by

$$\delta\lambda_j \cong - \left. \frac{\partial P / \partial \alpha_k}{\partial P / \partial z} \right|_{z=\lambda_j} \delta\alpha_k. \quad (10.32)$$

We can evaluate the partial derivatives in Eq. (10.32) from Eq. (10.29) and the equivalent form

$$P(z, \alpha_k) = (z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_n). \quad (10.33)$$

First, using Eq. (10.29), we compute

$$\left. \frac{\partial P}{\partial \alpha_k} \right|_{z=\lambda_j} = \lambda_j^{n-k}, \quad (10.34)$$

and next, using Eq. (10.33), we compute⁷

$$\left. \frac{\partial P}{\partial z} \right|_{z=\lambda_j} = \prod_{\ell \neq j} (\lambda_j - \lambda_\ell). \quad (10.35)$$

Thus Eq. (10.32) reduces to

$$\delta\lambda_j = - \frac{\lambda_j^{n-k}}{\prod_{\ell \neq j} (\lambda_j - \lambda_\ell)} \delta\alpha_k. \quad (10.36)$$

We can say things about root sensitivity by examining Eq. (10.36). The numerator term varies with the index number of the parameter whose variation we are considering. Because we are dealing with a stable system, the magnitude of λ_j is less than one, so the larger the power of λ_j^{n-k} the smaller the variation. We conclude that the most sensitive parameter is α_n , the constant term in Eq. (10.29). However, for values of λ_j near the unit circle, the relative sensitivity decreases slowly as k gets smaller. The denominator of Eq. (10.36) is the product of vectors from the characteristic roots to λ_j . This means that if all the roots are in a cluster, then the sensitivity is high, and if possible, the roots should be kept far apart. For example, if we wish to construct a digital low-pass filter with a narrow-pass band and sharp cutoff, then the system will have many poles in a cluster near $z = 1$. If we implement such a filter in the control canonical form (Fig. 4.8c), then the sensitivity given by Eq. (10.36) will have many factors in the denominator, all small. However, if we implement the *same* filter in the cascade or parallel forms, then the sensitivity factor will have only one term. Mantey (1968) studies these issues and quotes an example of a narrow-bandpass filter of six poles for which

⁷ This is true if $P(z)$ has only one root at λ_j . At a multiple root, this derivative is zero and the coefficient of $\delta\alpha_k$ in Eq. (10.36) is not bounded.

the parallel realization was less sensitive by a factor of 10^{-5} . In other words, it would take 17 bits of additional accuracy to implement this example in direct form over that required for the parallel or cascade form! The papers collected by Rabiner and Rader (1972) contain many other interesting results in this area.

◆ Example 10.5 *Illustration of Parameter Storage Errors*

Compare the sensitivity of a fourth-order controller in cascade form

$$D(z) = \frac{(z + 1)^3}{(z - 0.9)(z - 0.85)(z - 0.8)(z - 0.75)}$$

versus the same controller in direct form

$$D(z) = \frac{z^3 + 3z^2 + 3z + 1}{z^4 - 3.3000z^3 + 4.0775z^2 - 2.2358z + 0.4590}.$$

Solution. If the transfer function is realized as a cascade of the first-order terms, then it is clear that the pole nearest to the unit circle is at $z = 0.9$, and a change in the coefficient by 0.1 will move this pole to the unit circle and lead to instability. This is a percent change of $(0.1/0.9)100 = 11.1\%$. On the other hand, if the controller is realized in one of the direct forms—either control or observer canonical form—then the coefficients used are those in the polynomial form. In this case, by numerical experimentation it is found that a change of a_4 from 0.4590 to 0.4580, a root is moved to the unit circle. This is a change of only $(0.001/0.4590)100 = 0.22\%$. Thus we see that the cascade form is more robust to parameter changes by a factor of almost 50 over the direct forms!

10.3 Limit Cycles and Dither

As a final study of the effects of finite word length in the realization of digital filters and compensators we present a view of the quantizer as a signal-dependent gain and analyze more closely the motions permitted by this nonlinearity. One type of motion is an output that persists in spite of there being no input and that eventually becomes periodic: Such a motion is called a **limit cycle**.

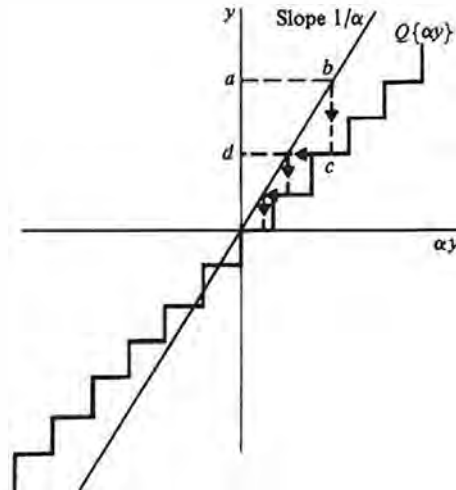
To analyze a nonlinear system we must develop new tools because superposition and transforms do not apply. One such tool is to use graphic methods to solve the equations. For example, suppose we have the first-order equation

$$y(k + 1) = Q\{\alpha y(k)\}. \quad (10.37)$$

If we plot y versus αy as in Fig. 10.6, we can also plot the function $Q(\alpha y)$ and trace the trajectory of the solution beginning at the point a on the y -axis. Across from a at point b , we plot the value αy from the line with slope $1/\alpha$. Below b at c is the quantization of αy , and hence the next value of y , shown as d . We

Figure 10.6

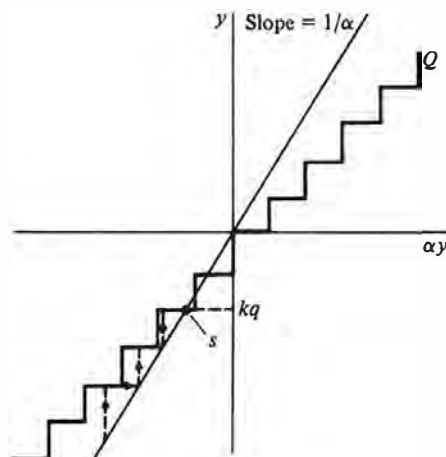
Trajectory of a first-order system with truncation quantization



thus conclude that the trajectory can be found by starting with a point on the $(1/\alpha)$ -line, dropping to the $Q\{\cdot\}$ staircase, projecting left to the $(1/\alpha)$ -line again, dropping again, and so on, as shown by the dashed lines and arrowheads. Note that the path will always end on the segment of zero amplitude where $\alpha y \leq q$. Now, however, suppose the initial value of y is negative. The trajectory is plotted in Fig. 10.7, where projection is up to Q and to the right to the $(1/\alpha)$ -line. Note that this time the trajectory gets stuck at the point s , and the response does *not* go to zero. The point s is an equilibrium or stationary point of the equation. If the initial value of y had a magnitude smaller than s , then the motion would have

Figure 10.7

Trajectory of a first-order system with truncation quantization and negative initial condition



moved down to the next equilibrium where the $(1/\alpha)$ -line and $Q\{\cdot\}$ intersect. It is clear that s represents the largest value of y that is at equilibrium and that this value depends on α . We should be able to find a relation between kq , the largest value of y at equilibrium, and α , the time constant of the filter. In fact, from inspection of Fig. 10.7 we see that the largest value of y at a stationary point is a value $y = -kq$ such that

$$\begin{aligned} -kq\alpha < -kq + q, & \quad \text{or} \quad k\alpha > k - 1, & \quad \text{or} \\ k < \frac{1}{1 - \alpha}, & \quad \text{or} \quad |y| < q \frac{1}{1 - \alpha}. \end{aligned} \quad (10.38)$$

The last result is the same as the worst-case bound that would be found from Eq. (10.8) with the maximum value of error q for truncation rather than $q/2$ as is the case for round-off. Thus we find in the first-order system that the worst case is rather likely to be realized.

In order to extrapolate these results to a higher-order system we must find a more subtle way to look at them. Such a viewpoint is provided by the observation that at the equilibrium points of Fig. 10.7 the gain of the quantizer is exactly $1/\alpha$. If we consider the quantizer as a variable gain, then the equilibrium occurs at those points where the combined gain of quantizer and parameter α are unity, which, for a linear system, would correspond to a pole at $z = 1$. The extrapolation of this idea to higher-order systems is to examine the range of possible **equivalent gains** of the quantizers and to conjecture that the limiting motion will be no larger than the largest signal for which the linear system with the resulting equivalent gain(s) has a pole on the unit circle.

We will illustrate the conjecture by means of the second-order control canonical form shown in Fig. 10.8(a) with the quantizer characteristic shown in Fig. 10.8(b) corresponding to round-off rather than truncation. Note from Fig. 10.8(b) that the staircase of this quantizer is centered about the line of slope 1.0 passing through the origin. The characteristic equation for this system, if quantization is ignored, is

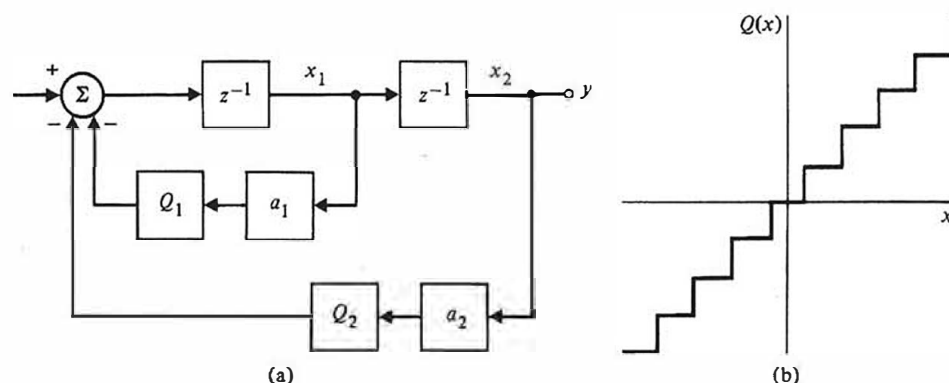
$$\begin{aligned} z^2 + a_1 z + a_2 &= 0, \\ z^2 - 2r \cos \theta z + r^2 &= 0, \end{aligned}$$

where we assume that $0 < a_2 < 1$ and a_1 corresponds to a real angle. Thus we see that the system will have complex roots on the unit circle only if the action of the quantizer is to make the effective value of $a_2 = r^2 = 1.0$. Thus the condition for an oscillation is that $Q(a_2 y) \geq y$. Following an analysis similar to Eq. (10.38), the property of the quantizer is that $Q(x - q/2) \leq q/2$. If we let $a_2 y = kq - q/2$, then the oscillation condition becomes

$$kq \geq \frac{kq - q/2}{a_2},$$

Figure 10.8

A second-order system in control canonical form and the quantizer characteristic corresponding to round-off



from which the amplitude is predicted to be less than

$$kq < \frac{1}{2} \frac{q}{1 - |a_2|}. \quad (10.39)$$

The effect of quantization on the a_1 -term influences only the frequency of the oscillations in this model. For this purpose, because the equivalent radius is 1, the equation for the digital frequency is $\theta = \cos^{-1}(-a_1/2)$.

dither

Another benefit of the view of quantization as a variable gain is the idea that a second signal of high frequency and constant amplitude added to the input of the quantizer can destroy the limit cycle. Such a signal is called a **dither**, and its purpose is to make the effective gain of the quantizer 1.0 rather than something greater than 1. Consider again the situation sketched in Fig. 10.7 with the signal stuck at s . If an outside high-frequency signal of amplitude $3q$ is added to y , then one can expect that although the output would contain a fluctuating component, the average value would drift toward zero rather than remain stuck at s . If the frequency of the dither is outside the desired pass band of the device, then the result is improved response; that is, a large constant bias or else a high-amplitude, low-frequency, self-sustained, limit-cycle oscillation can sometimes be removed this way at the cost of a low-amplitude, high-frequency noise that causes very low amplitude errors at the system output.

◆ Example 10.6 Quantization-Caused Limit Cycles and Effect of Dither

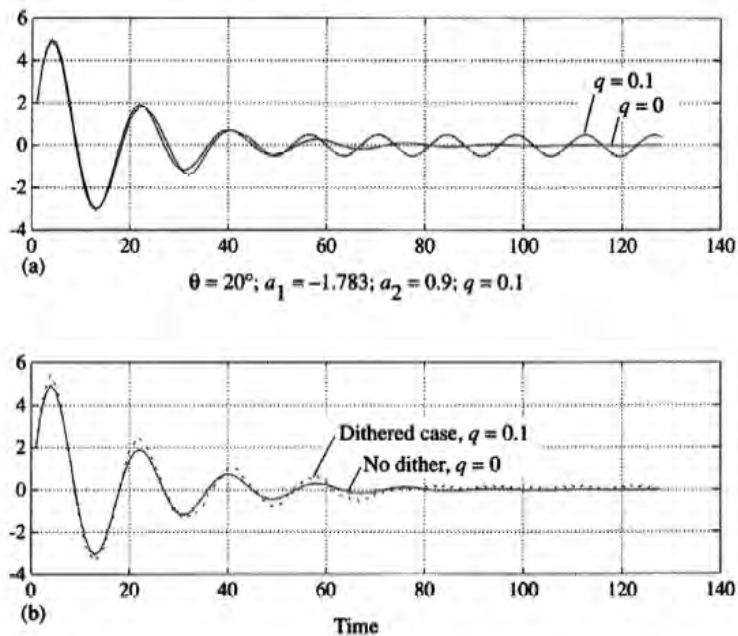
For the system in Fig. 10.8 with $a_1 = -1.78$ and $a_2 = 0.9$,

- (a) determine the response to the initial condition of $x_1 = 2$ and $x_2 = 0$ with $q = 0$ and $q = 0.1$. Compare the results to that predicted by Eq. (10.39).
 (b) Determine a dither signal that improves the response.

Solution. A simulation of the system of Fig. 10.8 with and without quantization is shown in Fig. 10.9(a). The limit cycle is clearly visible for the $q = 0.1$ case. The system parameters correspond to roots at a radius of 0.95 at an angle of 20° . The quantization level was set at 0.1. The response with quantization clearly shows the limit cycle with an amplitude of 0.5, which is exactly the value given by Eq. (10.39) with $a_2 = 0.9$. The period of the limit cycle can be seen from the figure to be approximately 14 samples. If we compute the frequency corresponding to the value of $a_1 = -1.78$ with the radius taken as 1 in order to reflect the fact that the system is oscillating, we find $\theta = \cos^{-1}(-a_1/2) = 27^\circ$. This angle corresponds to an oscillation with a period of 13.2 samples, an estimate that is quite good, considering that the true period must be an integral number of sample periods.

In Fig. 10.9(b) are plotted the response with $q = 0$ and the response with $q = 0.1$ with dither added. In this case, after some experimentation, it was found that a square-wave dither at the Nyquist frequency worked quite well at an amplitude of $4q$. In that case the steady-state response has been reduced in amplitude from 0.5 to 0.1 and the frequency has been increased from that having a period of 14 to that having a period of 2, namely the Nyquist frequency. The experiments found that dither of less amplitude did not remove the natural limit cycle and dither of higher amplitude gave a larger steady-state error response. A random dither was

Figure 10.9
 (a) Second-order response with and without quantization showing limit cycle.
 (b) Second-order response with quantization $q = 0.1$ and dither $= 4q$



also tried, which was found to be much less effective than the Nyquist frequency square wave. Unfortunately, the selection of the amplitude and signal shape of an effective dither remains more a matter for experimentation than theory.

10.4 Summary

- Multiplication round-off errors are bounded, in steady-state, according to

$$|\tilde{y}_{ss}(\infty)| \leq |H_1(1)| \frac{q}{2}, \quad (10.10)$$

where H_1 is the transfer function between the quantization error and the output; and $|H_1(1)|$ is the dc gain of H_1 .

- Multiplication round-off errors can be analyzed under most conditions by considering them to be an additional *random* input at each multiplication. The distribution of the random input is flat, its mean is 0, its variance is $R_x = q^2/12$, and it can be considered to be white (no time correlation).
- The effect of the random multiplication round-off error is analyzed by using the discrete Lyapunov equation

$$\mathbf{R}_x(\infty) = \Phi_1 \mathbf{R}_x(\infty) \Phi_1^T + \Gamma_1 R_w \Gamma_1^T, \quad (10.24)$$

which is evaluated via MATLAB using `dlyap.m` or `qrms.m`. For 16 and 32 bit computers, these errors are usually negligible. They could be significant using a computer with 8 bits or less, as will be shown in Fig. 11.6.

- The effect of parameter round-off (or parameter storage error) is systematic and has the capability to render an otherwise stable system unstable. It was shown that a parallel or cascade implementation of the difference equations significantly reduces the sensitivity to this type of error as does the use of a large number of bits to represent the parameters being stored.
- Under certain conditions (usually lightly damped systems), uncommanded oscillations will occur due to quantization called **limit cycles** which can be alleviated by the addition of low amplitude oscillations called **dither**.

10.5 Problems

- 10.1** For the system from Eq. (10.5) with $\alpha = 0.9$ and $q = .01$, what is the rms of the quantization error of y that you expect for random input, $u(k)$?
- 10.2** Verify Fig. 10.4 by simulating the response of the system from Eq. (10.5) to $u(k) = 1$ with and without quantization of the αy multiplication. Use $\alpha = 0.9$, $q = .02$, and compute the response from $k = 0$ to $k = 100$, repeating the simulation several times. Compute σ

of the difference between the case with and without quantization at $k = 5$ and $k = 95$ and compare with Fig. 10.4. Comment on the match you found and discuss any discrepancies.

10.3 A digital lowpass filter is described by the equation

$$y(k) = Q_2[\alpha y(k-1)] + Q_1[(1-\alpha)u(k-1)].$$

Assume that $\alpha > 0$ and that the input, $u(k)$, is a slowly varying signal that, for the purpose of this analysis, can be approximated by a constant. The magnitude of the input is restricted to be less than 0.99.

- If both quantizers operate with ℓ bits to the right of the point, what is the value of the quantum q for this case?
- Give an expression in terms of α for the minimum value that ℓ must have to guarantee that quantization error cannot cause y to exceed 1.0.
- Evaluate your expression in part (b) and give the necessary bit count if $\alpha = 0.9, 0.98, 0.995$.
- Suppose the input A/D quantizer Q_1 is fixed at 12 bits with 11 bits to the right of the fixed point. At what value of α can the quantization error alone cause the input to Q_2 to equal unity?

10.4 A digital filter with the structure of Fig. 10.8 is preceded by an A/D converter having 12 bits of accuracy, with 11 bits to the right of the fixed point. The quantizer Q_1 is a 16-bit word length scaled to have 3 bits to the left of the point. The quantizer Q_2 is scaled to have only 1 (sign) bit to the left of the point in a 16-bit word length. Let $a_1 = -1.6$ and $a_2 = 0.81$.

- Give the quantum sizes q_i for the converter and each internal round-off quantizer.
- Give the relevant transfer functions necessary to compute the output quantization error.
- Compute the steady-state worst error at the output due to quantization.
- Use MATLAB to compute the worst-case bound on output quantization error.
- Use MATLAB to compute the rms error due to quantization using the white-noise model.

10.5 For the second-order observer canonical form shown in Fig. 10.10,

- compute the transfer functions from the quantizers to the output, y . Note carefully how many need to be computed.
- If $b_1 = b_2 = 1$, $a_1 = -1.6$, $a_2 = 0.81$, what is the maximum steady-state error due to equal rounding quanta of $\pm q/2$?
- Show that the stochastic state-transition model to use Eq. (10.24) on this system has

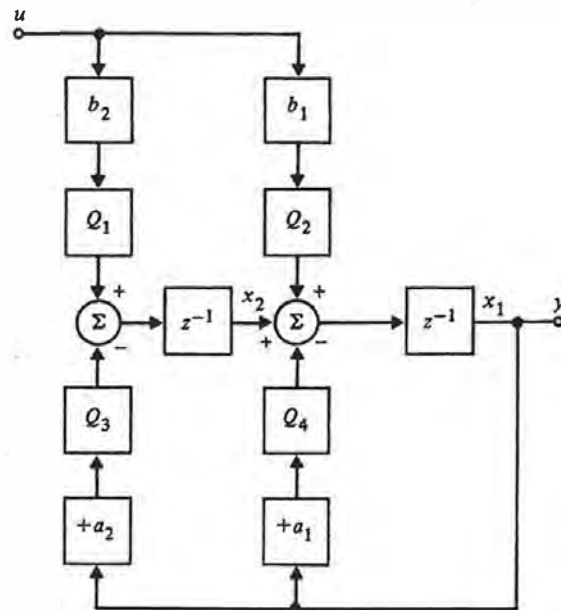
$$\Phi = \begin{bmatrix} 1.6 & 1 \\ -0.81 & 0 \end{bmatrix}, \quad \Gamma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R_w(\infty) = \frac{q^2}{6}, \quad \mathbf{H} = [1 \quad 0].$$

- Assume that

$$\mathbf{R} = \begin{bmatrix} a & b \\ b & c \end{bmatrix},$$

solve Eq. (10.24) for this case, and compute $R_y(0)$ from Eq. (10.27).

Figure 10.10
A second-order system
with quantization



- 10.6 Solve for the rms output due to round-off for the system of Fig. 10.8 using computer tools if $a_1 = -1.6$ and $a_2 = 0.81$.
- 10.7 The filter shown in Fig. 10.10 has input from an A/D converter that has 10 bits, 9 bits to the right of the fixed point. The input magnitude is restricted to 1. The four quantizers are based on 16-bit words scaled to have, respectively, 12, 12, 9, and 13 bits to the right of the points (i.e., Q_3 has 9 bits to the right). The parameters are $a_1 = -1.6$, $a_2 = 0.81$, and $b_1 = b_2 = 1$.
- Give the quanta q_i for each quantization location.
 - Give the respective transfer functions from each quantizer to the output.
 - Use MATLAB to compute the worst-case error bound on the output quantization error.
 - Use MATLAB to compute the rms output error for this system based on the white-noise model of quantization error.
- 10.8 Consider the discrete compensation whose transfer function is

$$D(z) = \frac{z^3}{(z - 0.5)(z - 0.55)(z - 0.6)(z - 0.65)}$$

- If implemented as a cascade of first-order filters with coefficients stored to finite accuracy, what is the smallest coefficient perturbation that could cause instability? Which parameter would be most sensitive?
- Draw an implementation of $D(z)$ in control canonical form with parameters a_1 , a_2 , a_3 , and a_4 . Compute the first-order sensitivities of these parameters according to Eq. (10.36). Which parameter is most sensitive? Compare the

sensitivities of root locations for the cascade and the control forms for these nominal root positions.

- (c) Using the root locus, find the maximum deviations possible for the parameters of the control canonical form and compare with the results of part (b).

10.9 For the equivalent-gain hypothesis of limit-cycle behavior,

- (a) Show that the equivalent gain of the round-off quantizer is given by

$$G_q \leq \frac{2k}{2k-1}, \quad k = 1, 2, \dots$$

for a signal of amplitude greater than or equal to $kq - q/2$.

- (b) What is the largest-amplitude limit cycle you would expect for a second-order control canonical form system with complex roots at a radius of 0.9 and a quantization level of 0.01?

10.10 For the system of Fig. 10.8,

- (a) Use MATLAB to simulate the response to the initial conditions $x_1 = 2$ and $x_2 = 0$ with zero input. Use $a_1 = -\sqrt{2}a_2$ and $a_2 = 0.8, 0.9, 0.95$, and 0.98 . Compare the amplitudes and frequencies of the limit cycles (if any) with the values predicted.
- (b) Add dither at the Nyquist frequency to the quantizer of $a_2 y$ with amplitude A and find the dither amplitude that minimizes the peak output in the steady state.

10.11 Repeat Example 10.6, but assume the design full scale is ± 20 units and the system is being implemented with an 8 bit cpu. Find the amplitude and frequency of the limit cycle, compare with the theory, and find a dither that improves the situation as much as possible. [Hint: Fig1009.M available from www.elliskagle.com will be useful.]

10.12 For the system of Example 10.6, assuming the design full scale is ± 20 units, find the number of bits required in the cpu to eliminate the limit cycle without any dither. [Hint: Fig1009.M available from www.elliskagle.com will be useful.]

10.13 For a system structured as in Example 10.6 with the same initial conditions, but with the coefficients selected so that the equivalent damping is $\zeta = 0.05$, assume the design full scale is ± 20 units and find the number of bits required in the cpu to eliminate the limit cycle without any dither. [Hint: Fig1009.m available from www.elliskagle.com will be useful.]

10.14 For the discrete system

$$D(z) = \frac{z^3 + 3z^2 + 3z + 1}{z^4 - 3.3000z^3 + 4.0775z^2 - 2.2358z + 0.4590},$$

how many bits are required for parameter storage in order to keep errors in the pole locations to be less than 0.01 in the z -plane in either the real or complex directions? Assume the system is to be implemented in the direct form.

♦ 11 ♦

Sample Rate Selection

A Perspective on Sample Rate Selection

The selection of the best sample rate (or rates) for a digital control system is a compromise. Generally, the performance of a digital controller improves with increasing sample rate, but cost may also increase with faster sampling. A decrease in sample rate means more time is available for the control calculations; hence slower computers are possible for a given control function, or more control capability is available for a given computer. Either result lowers the cost per function. For systems with A/D converters, slower sampling means less conversion speed is required, which will also lower cost. Furthermore, we will see that faster sampling can sometimes require a larger word size, which would also increase cost. All these arguments suggest that the best choice when considering the unit product cost is the slowest sample rate that meets all performance specifications.

On the other hand, digital controllers are often designed and built for systems where a very small number will be built. In this case, the cost of the design effort can be more than the unit product costs, and the savings in design time that are realized with very fast ($\geq 40 \times$ bandwidth) sampling dictates that a higher rate is the best choice.

This chapter will discuss the influence of the sample rate on system performance in order to give some insight into how to reconcile the issues for a given design.

Chapter Overview

Section 11.1 examines the fundamental limit on the sample rate imposed by the sampling theorem; Section 11.2 covers the effect on the time response, smoothness, and time lags; Section 11.3 examines the regulation effectiveness as measured by the response errors from random disturbances; Section 11.4 looks at the effect of sample rate on the sensitivity to plant parameter variations; and Section 11.5 examines how sample rate affects the error due to the measurement noise

and the influence of analog prefilters or antialiasing filters on this error. Since there are often conflicts in selecting the sample rate for different functions in a controller, some designers elect to have more than one sample rate (a “multirate system”) and this topic is discussed in Section 11.6.

11.1 The Sampling Theorem’s Limit

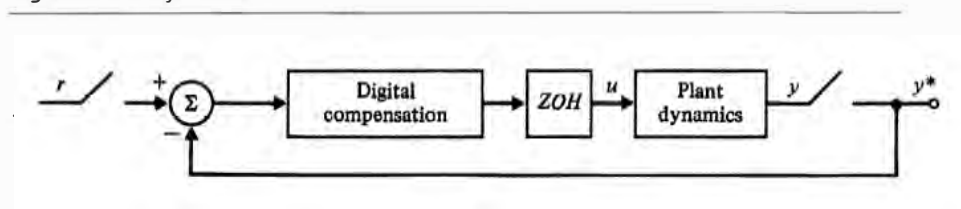
An absolute lower bound for the sample rate would be set if there is a specification to track certain command or reference input signals. This bound has a theoretical basis from the sampling theorem discussed in Section 5.2. Assuming we can represent our digital control system by a single-loop structure as depicted in Fig. 11.1, we can specify the tracking performance in terms of the frequency response from r to y . The sampling theorem states that in order to reconstruct an unknown band-limited continuous signal from samples of that signal, one must use a sample rate at least twice as fast as the highest frequency contained in the unknown signal. This theorem applies to a feedback controller like the one illustrated in Fig. 11.1, because r is an unknown signal that must be followed by the plant output y . If we want the system to track r up to a certain closed-loop bandwidth, ω_b , it follows that r will have spectral content up to that frequency. Based on the sampling theorem therefore, the sample rate must be at least twice the required closed-loop bandwidth of the system, that is

$$\frac{\omega_s}{\omega_b} > 2. \quad (11.1)$$

If a designer were to specify a certain value of the closed-loop bandwidth and then pick a sample rate that violated Eq. (11.1), the result would be that the sampled values of r would be aliased as discussed in Section 5.2, and the result would be a system response that was unstable or considerably slower than specified.

A similar argument can be made with regard to the closed-loop roots of a system, which typically would be slightly slower than the bandwidth. If a designer specified a closed-loop root by its s -plane location and failed to sample at twice its frequency, the actual root realized will be aliased and can have little

Figure 11.1
Digital control system schematic



resemblance to that specified. In practice, no designer would consider such a low sample rate; we bring it up only because it marks the theoretical lower limit of possibilities.

11.2 Time Response and Smoothness

Equation (11.1) provides the fundamental lower bound on the sample rate. In practice, however, this theoretical lower bound would be judged far too slow for an acceptable time response. For a system with a rise time on the order of 1 sec (which translates to a closed-loop bandwidth on the order of 0.5 Hz), it would be typical to choose a sample rate of 10 to 20 Hz in order to provide some smoothness in the response and to limit the magnitude of the control steps. This means that the desired sampling *multiple* ($= \omega_s/\omega_b$) for a reasonably smooth time response is

$$20 < \frac{\omega_s}{\omega_b} < 40. \quad (11.2)$$

◆ Example 11.1 Double Integrator Control Smoothness vs. Sample Rate

Compute the unit step response of the double integrator control problem as developed in Example 8.1 and plot the output x_1 , its rate x_2 , and control u time histories. Find the feedback gains for sample rates of 4, 8, 20, and 40 times the bandwidth, ω_b , so that the responses all have closed-loop roots in the z -plane that are equivalent to $s = 0.5\omega_b(1 \pm j)$. Discuss your results.

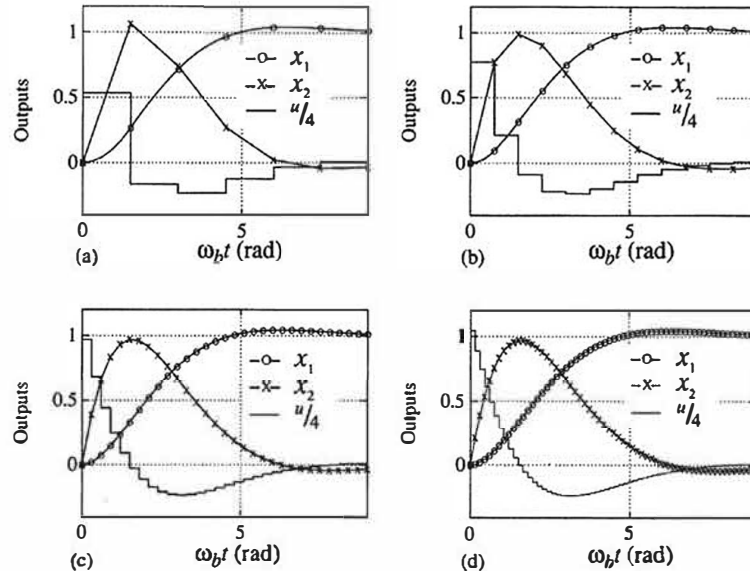
Solution. The four responses are shown in Fig. 11.2. The gains were computed using pole placement where the z -plane poles were computed using $z = e^{sT}$. Note that the relation $s = 0.5\omega_b(1 \pm j)$ is approximate in that the actual bandwidth of the closed-loop frequency response may be slightly different.

It is interesting to note that the x_1 response was smooth for all cases, including the one with $\omega_s/\omega_b = 4$; however, the acceleration had large discontinuities and would have had a strong tendency to excite any flexible modes and produce high stresses in the actuator and its surrounding structure. In turn, these acceleration steps produced noticeable changes of slope in the velocity. A sampling multiple of $\omega_s/\omega_b \geq 20$ appears necessary for a reasonable smoothness.

The degree of smoothness required in a particular design depends on the application and is highly subjective. The commands issued to an electric motor

Figure 11.2

Double integrator step response for the sampling multiple with ω_s/ω_b equal to (a) 4, (b) 8, (c) 20, and (d) 40



can have large discontinuities, whereas the commands issued to hydraulic actuators are best kept fairly smooth.¹ The tolerance to roughness in the response also depends on the application; for example, if a person is being affected by the controller, a smooth ride is likely desirable. An unmanned satellite controller can be rough; however, slow sampling contributes to increased pointing errors from disturbances, a topic that is discussed in the following section.

In addition to the smoothness issue, it is sometimes important to reduce the delay between a command input and the system response to the command input. A command input can occur at any time throughout a sample period; therefore, there can be a delay of up to a full sample period before the digital controller is aware of a change in the command input. All the responses in Fig. 11.2 assumed that the controller was aware of the command input at time zero and that, therefore, all the attitude responses, x_1 , responded in a similar fashion. For systems with human input commands where the system response is critical (such as fly-by-wire flight control), the time delay alone suggests that the sample period be kept to a small fraction of the rise time. A pilot flying an airplane with digital fly-by-wire flight control will complain if the sampling delay is on the order of a tenth of a second from input action to the beginning of the response. Assuming we wish to keep

time delay

¹ Sometimes lowpass filters are placed between the ZOH output and the actuators to soften the discontinuities, but the filters must be taken into account during design and compensated for.

the time delay to be 10% of the rise time, a 10-Hz sample frequency should be used for 1 sec rise time or, in terms of the nondimensional sampling multiple

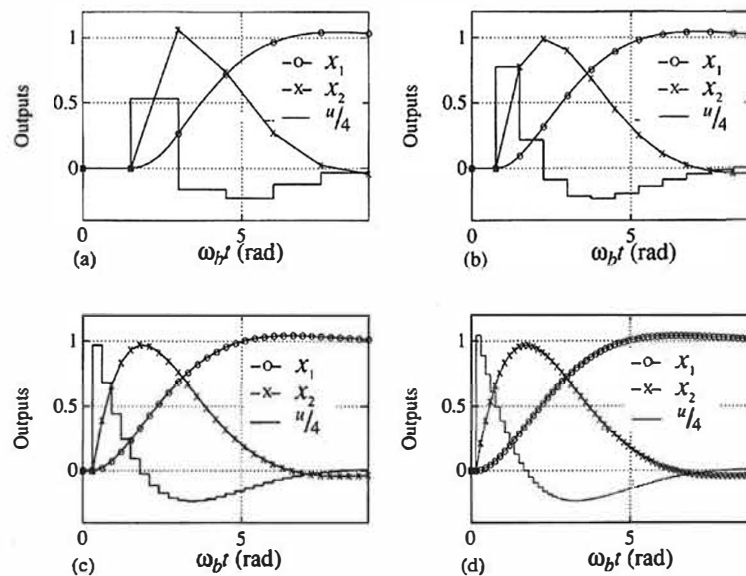
$$\frac{\omega_s}{\omega_b} \geq 20. \quad (11.3)$$

◆ Example 11.2 Double Integrator Response vs. Sample Rate

Repeat Example 11.1, but add a one cycle delay between the input command and the start of the control input in order to assess the worst case phasing of the input. Discuss the impact of this issue.

Solution. The result is shown in Fig. 11.3. It demonstrates the noticeable impact on the response for the two slower cases. The overshoot is unchanged because the controller was adjusted in each case to maintain the same z -plane roots, but the extra delay affected the rise time substantially as measured from the instant of the input command.

Figure 11.3
Double integrator step response with worst case phasing between command input and the sampler with ω_s/ω_b equal to (a) 4, (b) 8, (c) 20, (d) 40



11.3 Errors Due to Random Plant Disturbances

Disturbance rejection is an important aspect of any control system and, in many cases, is the most important one. In fact, Berman and Gran (1974) suggest that the sample rate for aircraft autopilots should be selected primarily on the basis of its effect on disturbance rejection. Indeed, this is the case for many control applications, although there are multitudes of applications with widely varying conditions where other factors are more important.

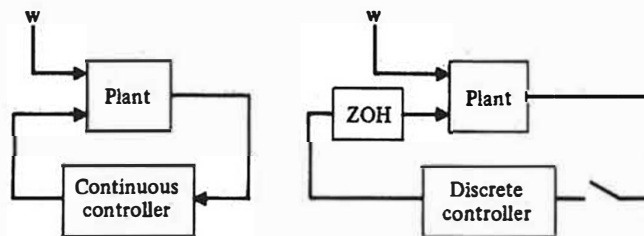
Disturbances enter a system with various characteristics ranging from steps to white noise. For determining the sample rate, the higher frequency random disturbances are the most influential; therefore, we will concentrate on their effect. In other words, we will look at disturbances that are fast compared to the plant and the sample rate, that is, where the plant noise can be considered to be white.

The ability of the control system to reject disturbances with a good continuous controller represents a lower bound on the magnitude of the error response that can be hoped for when implementing the controller digitally. In fact, some degradation over the continuous design must occur because the sampled values are slightly out of date at all times except at the very moment of sampling. In order to analyze the degradation of the digital controller as compared to the continuous controller, it is important to consider the effect of the noise [w in Eq. (4.45)] consistently, with both the continuous and the digital controllers.

The block diagram in Fig. 11.4 shows the situation. The plant noise, generally a vector quantity, w , is continuous in nature and acts on the continuous part of the system independently of whether the controller is continuous or discrete. Furthermore, we are at liberty to analyze the effect of w with a certain power spectral density regardless of what values of R_w or R_y were used in computing the estimator gains, L , and regardless of whether an estimator was used. Proceeding then, we have a continuous system represented by

$$\dot{x} = Fx + Gu + G_1 w, \quad (11.4)$$

Figure 11.4
Block diagrams of the
systems for disturbance
analysis



where the power spectral density of $\mathbf{w}$ is $\mathbf{R}_{\text{wpsd}}$ (alternatively referred to as the “white-noise intensity” or “mean-square spectral density”) so that the covariance of $\mathbf{w}$ is

$$E[\mathbf{w}(t)\mathbf{w}^T(t + \tau)] = \mathbf{R}_{\text{wpsd}}\delta(\tau).$$

The steady-state value of the covariance of $\mathbf{x}$ is given by the Lyapunov equation²

$$\mathbf{F}\mathbf{X} + \mathbf{X}\mathbf{F}^T + \mathbf{G}_1\mathbf{R}_{\text{wpsd}}\mathbf{G}_1^T = 0. \quad (11.5)$$

The solution to this equation, $\mathbf{X}$, ($= E[\mathbf{x}(t)\mathbf{x}(t)^T]$) represents the amplitude of the random response of the state due to the excitation from $\mathbf{w}$. It will be used to establish a baseline against which discrete controllers are compared. Note that the system matrices, $\mathbf{F}$ and $\mathbf{G}$, can represent a closed-loop system including a continuous controller. The solution is obtained by `lyap.m` in MATLAB.

It is also necessary to evaluate $\mathbf{X}$ when the system has a digital controller for the identical excitation applied to the plant. In order to do this, the entire system must be transferred to its discrete equivalent as given by Eq. (9.81) and discussed in Section 4.3. The discrete equivalent of Eq. (11.5) is given by Eqs. (9.90), (10.24), and (D.34) in slightly different contexts. The desired result for our purposes here, called the discrete Lyapunov equation (see `dlyap.m` in MATLAB) is

$$\Phi\mathbf{X}\Phi^T + \mathbf{C}_d = \mathbf{X}, \quad (11.6)$$

where

$$\mathbf{C}_d = \int_0^T \Phi(\tau)\mathbf{G}_1\mathbf{R}_{\text{wpsd}}\mathbf{G}_1^T\Phi^T(\tau)d\tau. \quad (11.7)$$

As discussed in Section 9.4.4, this integral can be approximated if T is shorter than all system time constants by

$$\mathbf{C}_d \cong \mathbf{\Gamma}_1\mathbf{R}_w\mathbf{\Gamma}_1^T, \quad \text{where} \quad \mathbf{R}_w = \frac{\mathbf{R}_{\text{wpsd}}}{T}. \quad (11.8)$$

When the approximation is not valid, it is necessary to evaluate Eq. (11.7) exactly using Van Loan's (1978) algorithm, which can easily be done using `disrv.m` in the Digital Control Toolbox. Therefore, in order to evaluate the effect of sample rate on the performance of a controller in the presence of white plant disturbances, we first evaluate Eq. (11.5) to find the baseline covariance ($\mathbf{X}$) and then repeatedly evaluate Eq. (11.6) with varying sample rates to establish the degradation versus sampling. It usually suffices to examine the diagonal elements for a performance measure and to compute their square roots to find the rms value, the quantity that is typically measured.

² See Kwakernaak and Sivan (1972).

◆ **Example 11.3** *Double Integrator Disturbance Response vs. Sample Rate*

Examine the effect of sample rate on the performance of a digital control system compared to a continuous control system for the double integrator plant used in Examples 11.1 and 11.2.

- (a) Assume the plant is driven by white noise entering in the same way as the control input with $R_{wpsd} = 1$ and assume the use of full-state feedback.
- (b) Assume the same plant noise, but now use an estimator with only x_1 measured with additive noise with $R_v = 1$. Repeat this case assuming there is quantization in the estimator equivalent to 7, 8, and 9 bit word size.

Solution.

- (a) The open loop F , G , and H are given by Eq. (4.47). The control gain, K , was determined by selecting optimal weighting matrices so that the closed-loop system had roots at $s = 0.5 \omega_b (1 \pm j)$ ($\zeta = 0.7$) and we assume that $\omega_b = \sqrt{2} \omega_n$. With full-state feedback the closed-loop continuous system matrix is given by $F_c = F - GK$. The plant disturbance noise enters the plant in precisely the same way that the control does, that is, $G_1 = G$. Therefore the response of a hypothetical continuous controller is found by solving Eq. (11.5) with F replaced with F_c and G_1 as stated. Because we wish to illustrate only the degradation of the discrete controller compared to the continuous one, the choice of $R_{wpsd} = 1$ has no effect on the results.

The family of discrete controllers with different sample periods were all designed to the same continuous cost function according to the method of Section 9.3.5, the idea being that all the discrete controllers should be trying to do the same thing. Each discrete design resulted in a unique K and system matrix $\Phi_c (= \Phi - T K)$, which was used in Eq. (11.6) to evaluate X . Because this example had no dynamic characteristics that were faster than the slowest sample period, the approximation for the plant noise given by Eq. (11.8) could have been used, although the exact calculation of Eq. (11.7) was actually used. The result shown in Fig. 11.5 is the ratio of the rms values for the discrete case to the continuous case. The specific curve shown is for the rms of the x_1 variable; however, both rms ratios are essentially identical.

If white plant disturbances were the dominant source of error in the system, one could conclude from this example that a sampling multiple of

$$\frac{\omega_s}{\omega_b} \cong 20 \quad (11.9)$$

would be a good choice. The relative errors grow quickly when sampling slower than this multiple, whereas the gain by sampling faster is simply to reduce the degradation from about 20% downward.

- (b) We can analyze the disturbance response of the system when the controller includes an estimator in a similar manner to that above. In this case, the system matrix Φ in Eq. (11.6) is defined by Eq. (8.62). Note that Φ now includes the plant dynamics as well as the estimator dynamics. The continuous control roots are in the same location as for part (a) and the continuous estimator roots were selected twice as fast as the control roots. The family of discrete controllers were found by using pole placement so that the discrete control and estimator roots were related to the continuous roots by $z = e^{sT}$. The lowest